In Power BI, data sources are categorized into different types based on their origin and how Power BI connects to them. Below is a detailed list of the major data sources in Power BI:

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1. File-Based Data Sources

Power BI can connect to various file types stored on a local machine or cloud storage.

Common File Formats:

- **Excel (.xlsx, .xlsm)** – Tables, pivot tables, Power Query, Power Pivot models.
- **CSV (.csv)** – Plain text with comma-separated values.
- **XML (.xml)** – Structured hierarchical data.
- **JSON (.json)** – Used for web APIs and structured data exchange.
- **PDF (.pdf)** – Extracts data from tables within PDFs.
- **Parquet (.parquet)** – Optimized columnar storage format used for big data.

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2. Database Data Sources

Power BI supports direct connections to databases, allowing for scheduled or live data retrieval.

Relational Databases:

- **SQL Server** – Microsoft's enterprise database.
- **Azure SQL Database** – Cloud-based SQL database.
- **Azure Synapse Analytics** – Scalable cloud data warehouse.
- **MySQL** – Open-source relational database.
- **PostgreSQL** – Advanced open-source relational database.
- **Oracle Database** – Enterprise-level database system.
- **IBM Db2** – Relational database from IBM.
- **Teradata** – High-performance analytics database.
- **SAP HANA** – In-memory computing database.
- **SAP Business Warehouse** – Enterprise data warehouse for SAP applications.

NoSQL & Big Data Sources:

- **Cosmos DB** – Microsoft's globally distributed NoSQL database.
- **MongoDB (via ODBC/connector)** – NoSQL database for unstructured data.
- **Amazon Redshift** – Cloud-based data warehouse service.
- **Google BigQuery** – Cloud data warehouse for analytics.

- **Snowflake** – Cloud-based data warehouse.

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3. Online Services & Cloud-Based Data Sources

Power BI integrates with various online services for business intelligence and analytics.

- **SharePoint Online & On-Premises** – Pulls data from lists, libraries, and files.
- **OneDrive & OneDrive for Business** – Connects to stored Excel or CSV files.
- **Google Drive** – Accesses cloud files via third-party connectors.
- **Azure Data Lake Storage** – Scalable cloud storage for big data.
- **Azure Blob Storage** – Stores unstructured data such as text, images, and logs.
- **Microsoft Dynamics 365** – Retrieves ERP and CRM data.
- **Salesforce** – Pulls data from Salesforce CRM.
- **Google Analytics** – Fetches website performance data.
- **Facebook, Twitter, LinkedIn** – Connects via APIs for social media analytics.
- **Adobe Analytics** – Web analytics platform.
- **Mailchimp** – Email marketing and campaign data.

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4. Web-Based & REST API Sources

Power BI can fetch data from web-based sources and APIs.

- **Web URL (Web Connector)** – Retrieves data from a public or private website.
- **REST APIs** – Custom API connections using JSON/XML.
- **OData Feed** – Web-based open data format used by various services.
- **Microsoft Graph API** – Accesses Microsoft 365 data.

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5. Other Data Sources

Power BI supports additional data sources via direct connections and third-party connectors.

- **ODBC (Open Database Connectivity)** – General connector for databases.
- **OLE DB (Object Linking and Embedding, Database)** – Used for legacy databases.
- **R & Python Scripts** – Import and manipulate data using scripting languages.
- **Spark** – Connects to Apache Spark clusters for big data processing.
- **Hadoop (HDFS)** – Connects to big data stored in Hadoop Distributed File System.
- **Azure Analysis Services** – Connects to enterprise-level data models.

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6. Real-Time & Streaming Data Sources

For real-time analytics, Power BI can connect to live data streams.

- **Azure Stream Analytics** – Processes real-time streaming data.
- **PubNub** – Real-time data messaging.
- **Power BI REST API** – Pushes data into Power BI datasets for real-time updates.

Connection Modes in Power BI

Depending on the source, Power BI offers different connection methods:

1. **Import Mode** – Loads data into Power BI's internal model for fast performance.
2. **DirectQuery** – Queries data directly from the source without importing it.
3. **Live Connection** – Used for connecting to SQL Server Analysis Services or Azure Analysis Services without importing data.
4. **Push Dataset** – Allows real-time data streaming.

Conclusion

Power BI provides a wide range of data source connectors, making it a powerful tool for business intelligence. You can integrate data from files, databases, cloud services, APIs, and real-time streams to create reports and dashboards.

Would you like help setting up a specific data source? 🚀